Harsh Nair

Embedded Firmware Engineer

RESUME_1343

Location: Warangal, Telangana, India | Email: harsh.nair.resume_1343@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 6 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 6 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: UART, ARM Cortex-M, Embedded Linux, Oscilloscope, FreeRTOS, C, SPI, STM32, I2C, Firmware Debugging, Git, C++

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Senior Embedded Firmware Engineer | KoraLink Innovations | Jan 2023 - Present

- Validated 731 hardware test vectors across protocol and timing corner cases
- Lowered field-reset incidents by 16% with watchdog recovery and diagnostics
- Improved battery runtime by 34% with peripheral and sleep-state optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Embedded Firmware Engineer | Asterline Engineering | Jul 2020 - Dec 2022

- Improved battery runtime by 31% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 14% using interrupt priority and queue tuning

PROJECTS

Motor Control Interface | Oscilloscope, Embedded Linux, Git, I2C

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Industrial Sensor Firmware | Git, C, Oscilloscope, I2C

- Built FreeRTOS tasks for calibrated sampling and watchdog recovery.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Meridian Technical University, 2020
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

ARM Cortex-M Workshop (2020)